

Coverage Report: Federal School Roster vs. the NU Master List

Date: 2026-07-27 · Artifacts: `csv_exports/nu_list_name_crosswalk_2026-07-27.csv` (full correspondence table), `csv_exports/nu_list_missing_schools_2026-07-27.csv` (gap list)

1. Headline coverage

After the full matching pipeline (CEEb linkage + name-variant rules + geographic audit), the two datasets align as follows:

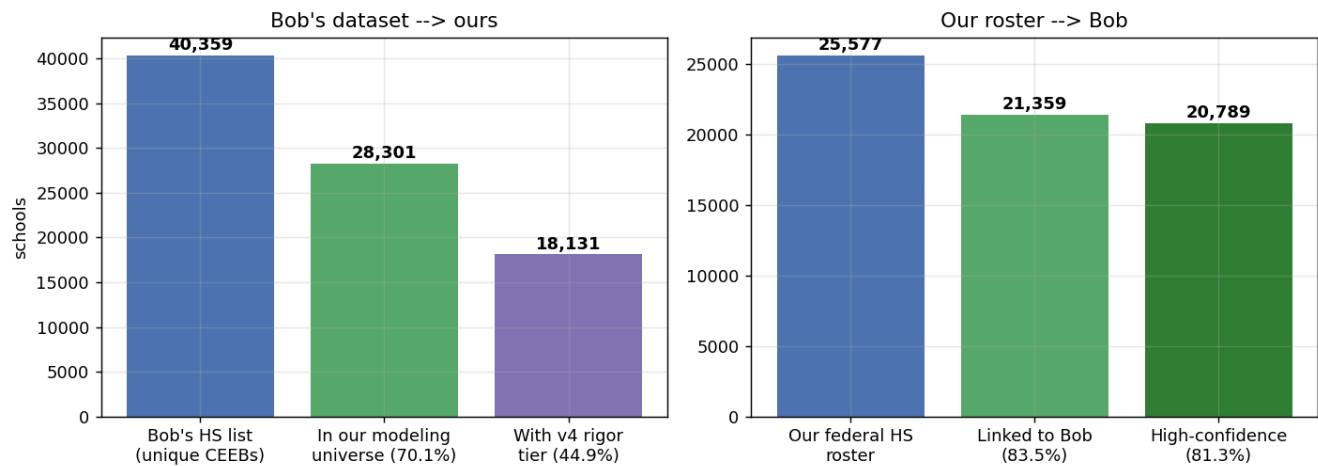


Figure 1: Bidirectional coverage

the NU master list → ours. Of the 40,359 unique high schools (by CEEB) in the NU org export, 28,301 (70.1%) appear in our federally-anchored modeling universe, and 18,131 (44.9%) carry a v4 rigor tier. The 30% that do not enter the universe are dominated by adult schools, vocational/training centers, credit-recovery and independent-study programs, and closed schools — only 2.8% of them were ever visited by NU and only 16% carry any NU analytics, i.e., they are largely dormant entries rather than active recruiting targets.

Our roster → the NU master list. Of our 25,577 verified public and private high schools (grades 9–12, enrollment ≥ 30), 21,359 (83.5%) link to a NU-list record — 20,789 (81.3%) at high confidence — leaving 4,218 (16.5%) genuinely absent from the NU master list.

2. Metric definitions

Metric	Definition
Modeling universe	Public + private high schools serving grades 9–12 with enrollment ≥ 30 , after removing colleges, closed schools, and non-school organizations
v4 rigor tier	Five-tier curricular-rigor classification (qualifying-density AP metric + verified IB), covering schools with ≥ 1 scoring component
High-confidence match	CEEb-exact, exact normalized name, name-subset with same city, or fuzzy ≥ 90 with same city
Review-tier match	Name-subset without city confirmation (570 rows) — usable but flagged for spot-checking

Metric	Definition
Missing school	A school in our federal roster with no NU-list record under any rule

3. Matching methodology

Matching proceeds through a five-tier funnel; every school receives an auditable rule label:

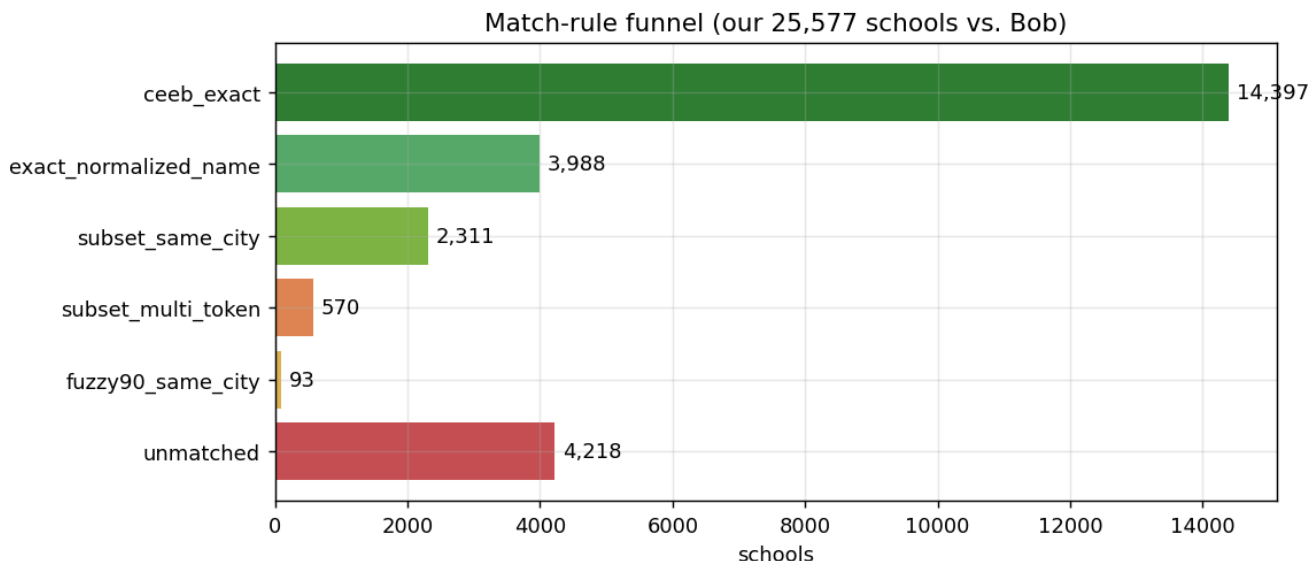


Figure 2: Match rules

1. **ceeb_exact (14,397)**. The school’s adjudicated CEEB code appears in the NU master list. CEEBs come from the NCES↔CEEBS junction after LLM adjudication of all review-tier pairs.
2. **exact_normalized_name (3,988)**. Token-normalized names match exactly within the state. Normalization strips generic words (High/School/HS/SHS/ “H S” /Senior/Junior/Secondary), expands abbreviations (Twp→Township, St→Saint, Sec→Secondary, Ft→Fort...), drops middle initials, and repairs the NU master list’s ~30-character name truncation (e.g., “...Johnso” → “Johnson”) when the cut token uniquely prefixes one of ours.
3. **subset_same_city (2,311)**. One name’s distinctive tokens are a subset of the other’s and the cities agree — this captures district prefixes (“Alief Hastings” ↔ “Hastings”) and person-name prefixes (“Patricia E Paetow” ↔ “Paetow” , “James W Robinson Jr Sec School” ↔ “Robinson Secondary”).
4. **fuzzy90_same_city (93)**. Token-set similarity ≥ 90 with the same city — catches misspellings and punctuation variants.
5. **subset_multi_token (570, review)**. Subset match without city confirmation; kept but flagged, since spot-checks found occasional false pairs.

A **geographic audit** validates the pipeline: for 15,263 matched public schools with coordinates in both datasets, the median NU-vs-NCES coordinate discrepancy is **1.0 km** and 94.3% lie within 10 km. The 364 pairs more than 50 km apart form a standing QA list of suspect matches (overlapping the CEEB digit-corruption issue documented in the EDA report).

4. The 4,218 schools missing from the NU master list

Composition. 3,979 public / 239 private; 147 schools with $\geq 1,000$ students and 274 more with 500–999; 3,149 under 200 students. Texas (430), Minnesota (381), and California (358) lead. The list includes **41 IB schools**, and 820 rows already carry a v4 rigor tier — the admissions team can triage them immediately.

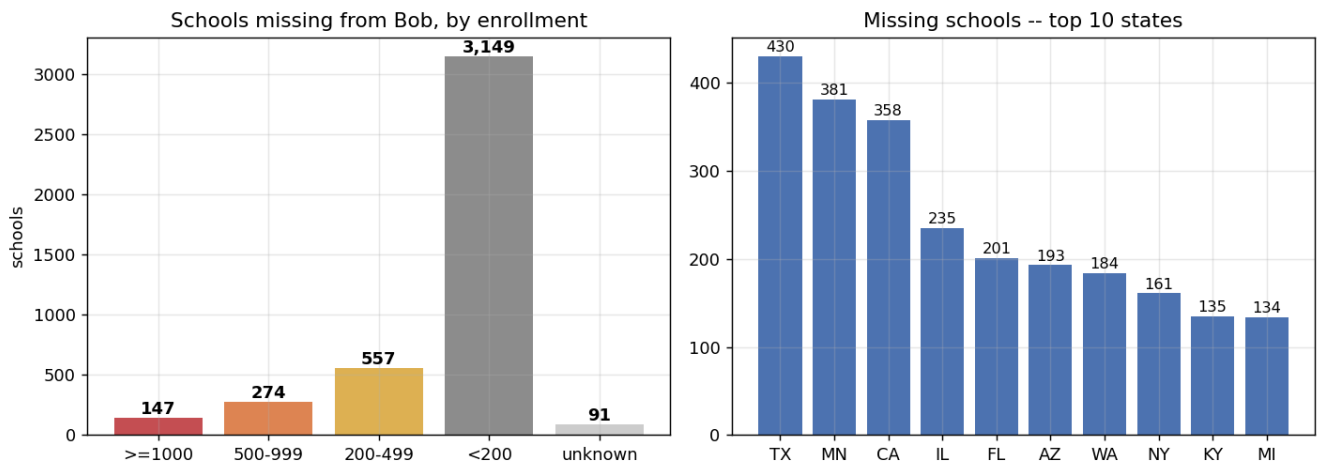


Figure 3: Missing profile

Why they are missing — three causes, in order of importance:

1. **A 2000s-cohort blind spot (confirmed).** Geographic audit of the largest missing schools proved five mega-schools (2,861–4,291 students) genuinely absent while their same-district neighbors are present: John A. Ferguson (Miami, opened 2003), Timber Creek (Orlando, 2001), Newsome (Lithia FL, 2003), Cherokee Trail (Aurora CO, 2003), Cypress Ridge (Houston, 2002). All opened in the early-2000s suburban construction wave — the NU master list appears never to have systematically back-filled that cohort.
2. **Small, rural, and newly opened schools (the long tail).** Median enrollment on the missing list is 39; these schools sit outside NU’ s historical recruiting footprint, so no one ever created records for them.
3. **Non-traditional formats.** Virtual academies, alternative programs, and micro-charters that issue diplomas but rarely interact with selective admissions.

Two rows are annotated `likely_in_nu_list_manual_review` (Johnson HS San Antonio $\leftrightarrow$ “Claudia Taylor Ladybird Johnso[n]” ; NV Learning Academy $\leftrightarrow$ a co-located “Academy Individualized Study”) — cases our conservative rules deliberately decline to auto-accept.

5. Recommendations

1. Feed `nu_list_name_crosswalk` into the pipeline’ s org join to replace pure-CEEB linkage (+6,962 links available immediately).
2. Hand the admissions team the missing-schools list sorted by enrollment: the 421 schools ≥ 500 students are direct recruiting-visibility additions; the long tail evidences platform completeness.
3. Investigate the five confirmed 2000s-cohort absences at the source system — if the ingest cutoff is real, more of that cohort is silently missing beyond our federal roster’ s reach.
4. Work the 364-pair geo-discrepancy QA list together with the CEEB digit-shift corruption flags.